



Data mining and analytics

Lec.8.

Data warehouse

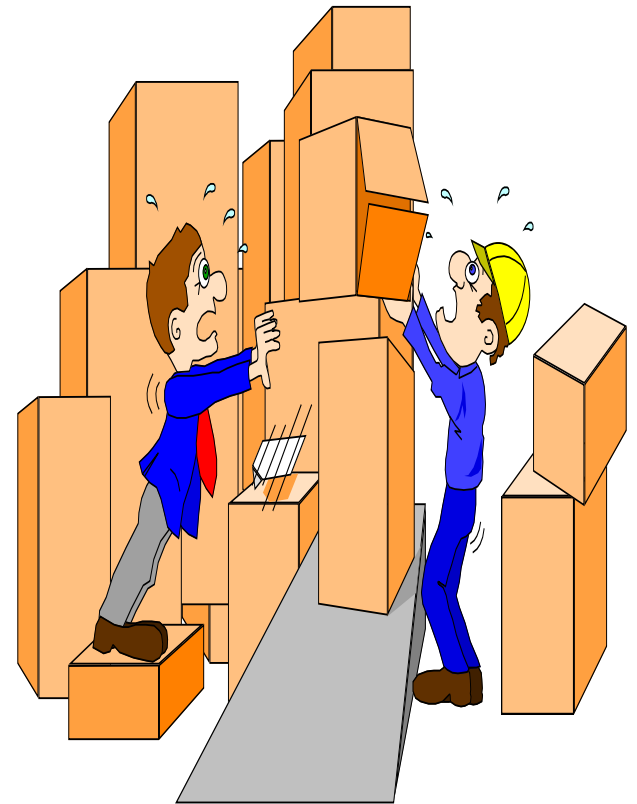
Data, Data everywhere yet ...



- I can't find the data I need
 - data is scattered over the network
 - many versions, subtle differences
- I can't get the data I need
 - need an expert to get the data
- I can't understand the data I found
 - available data poorly documented
- I can't use the data I found
 - results are unexpected
 - data needs to be transformed from one form to other

What are the users saying...

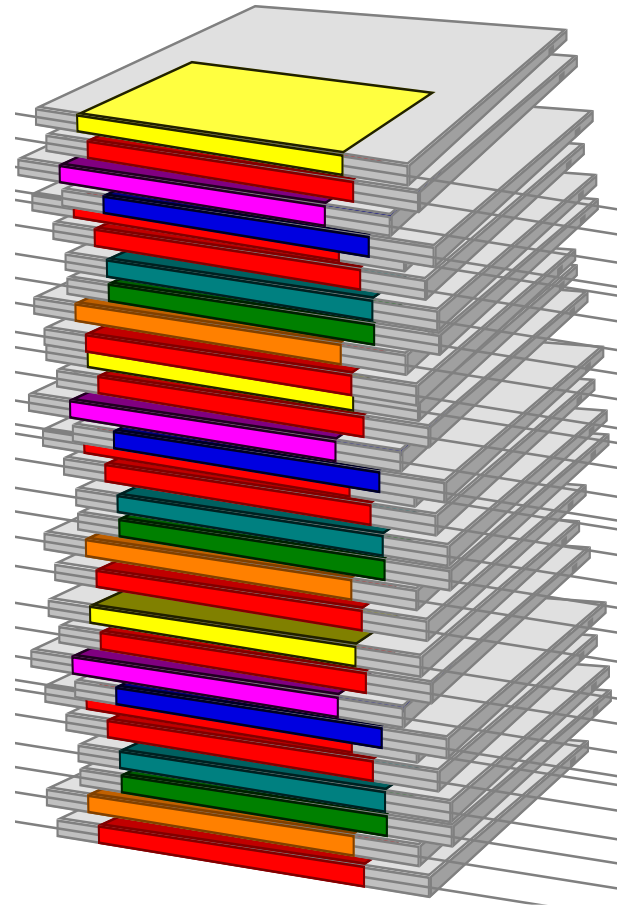
- Data should be integrated across the enterprise
- Summary data has a real value to the organization
- Historical data holds the key to understanding data over time
- What-if capabilities are required



What is a Data Warehouse?

Bill Inmon, Building the Data Warehouse 1996

A single, complete and consistent store of data obtained from a variety of different sources made available to end users in a way that they can understand and use in a business context.



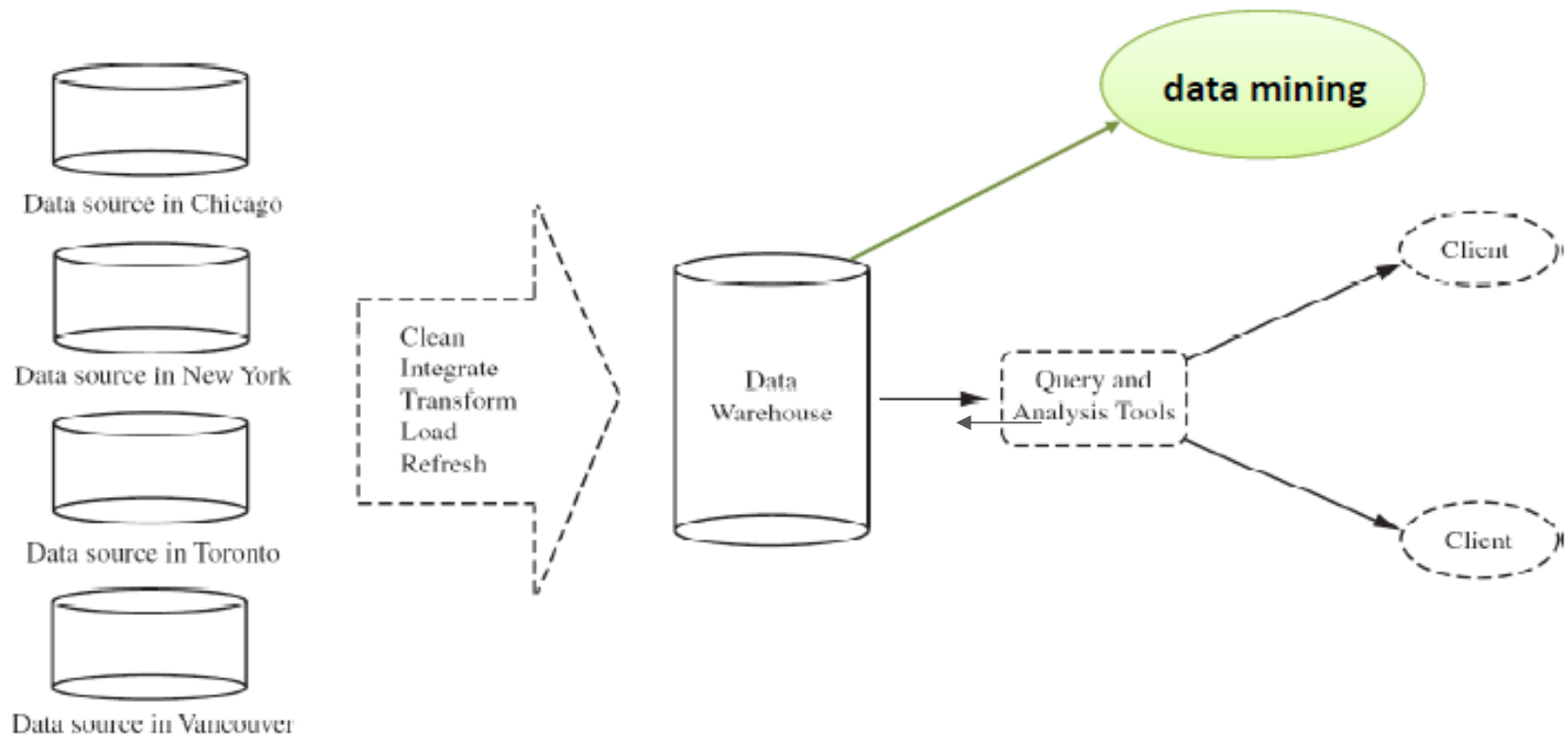
What is Data Warehouse?

A data warehouse is a collection of integrated databases designed to support a DSS.

A data warehouse is a collection of data that is used primarily in organizational decision making.

A data warehouse is a subject-oriented, integrated, time-variant, and nonvolatile collection of data in support of management's decision-making process.

Data Warehouse Framework



What is Data Warehousing?



A **process** of transforming **data** into **information** and making it available to users in a timely enough manner to make a difference

Data Warehousing It is a --



- Technique for assembling and managing data from various sources for the purpose of answering business questions. Thus making decisions that were not previously possible
- A decision support database maintained separately from the organization's operational database

- It needs a successful technician.
- Flexible.
- Team player.
- Good balance of business and technical understanding.

Characteristics of Data Warehouse

1. Subject-oriented:

- Data warehouse organized around subjects such as sales, product and customer.
- Data are organized based on how the users refer to them.
- It focuses on modeling and analysis of data for decision makers.
- Provide a simple and concise view around particular subject issues by excluding data that are not useful in the decision support process.

2. Integration

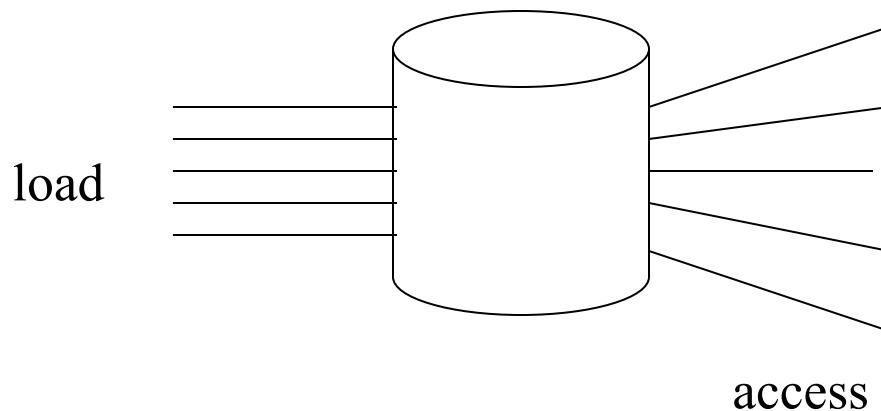
- Data Warehouse is constructed by integrating multiple heterogeneous sources.
- Data Preprocessing are applied to ensure consistency.
- All inconsistencies regarding naming convention and value representations are removed.

3. Time-variant

- The time horizon for the data warehouse is significantly longer than that of operational systems
 - Operational database: current value data
 - Data warehouse data: provide information from a historical perspective (e.g., past 5-10 years)
- Every key structure in the data warehouse
 - Contains an element of time, explicitly or implicitly
 - But the key of operational data may or may not contain “time element”

4. Nonvolatile

- Data once recorded cannot be updated. Data are stored in read-only format and do not change over time
- Data warehouse requires two operations in data accessing
 - Initial loading of data
 - Access of data



Data Warehouse environment:

- Contain:
 - The data store
 - Data mart
 - Metadata.

1.The Data Store

- Is the most common component of DW environment.
- Its function is to feed the data warehouse with data for the purpose of analysis.
- Data store is generally subject oriented, volatile and current commonly focused on customers, products, orders, policies, claims, etc...

2.The Data Mart

- A data mart contains a subset of corporate-wide data that is of value to a specific group of users. The scope is confined to specific selected subjects. For example, a marketing data mart may confine its subjects to customer, item, and sales. The data contained in data marts tend to be summarized.
- Data marts are usually implemented on low-cost. The implementation cycle of a data mart is more likely to be measured in weeks rather than months or years.

3. Metadata

- Last component of DW environments.
- It is information that is kept about the warehouse rather than information kept within the warehouse.
- Legacy systems generally don't keep a record of characteristics of the data (such as what pieces of data exist and where they are located).
- The metadata is simply data about data.

Metadata

- The name suggests some high-level technological concept, but it really is fairly simple.
- With the emergence of the data warehouse as a decision support structure, the metadata are considered as much a resource as the business data they describe.
- Metadata are abstractions -- they are high level data that provide concise descriptions of lower-level data.

Metadata

For example, a line in a sales database may contain:

4056 KJ596 223.45

This is mostly meaningless until we consult the metadata that tells us it was store number 4056, product KJ596 and sales of \$223.45

The metadata are essential ingredients in the transformation of raw data into knowledge. They are the “keys” that allow us to handle the raw data.

- **Technical Meta Data:** This kind of Metadata contains information about warehouse which is used by Data warehouse designers and administrators.
- **Business Meta Data:** This kind of Metadata contains detail that gives end-users a way easy to understand information stored in the data warehouse.

Components of the Metadata

- Transformation maps – records that show what transformations were applied
- Extraction & relationship history – records that show what data was analyzed
- Algorithms for summarization – methods available for aggregating and summarizing
- Data ownership – records that show origin
- Patterns of access – records that show what data are accessed and how often

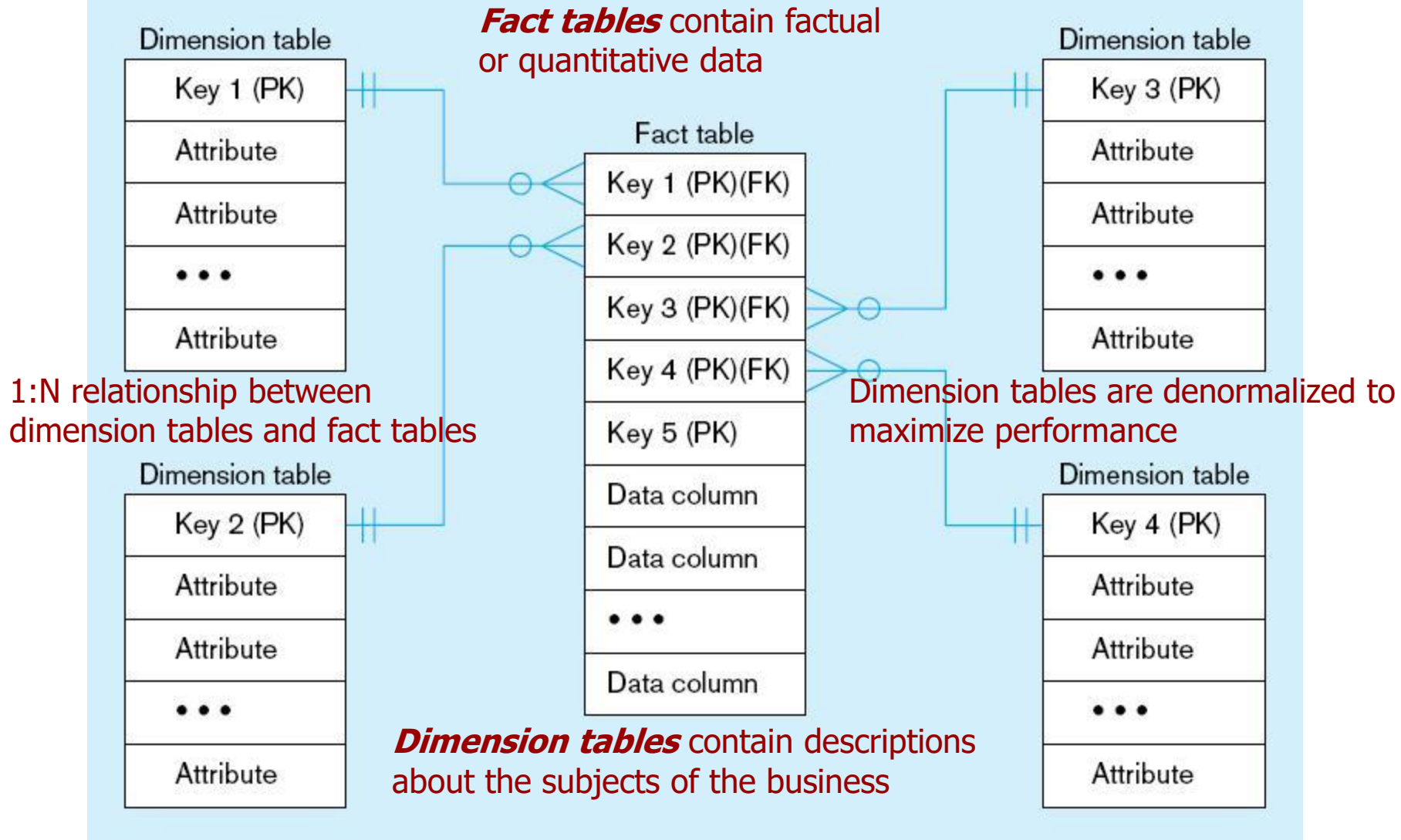
Data Warehouse Schema

- Star Schema
- Snowflake Schema
- Fact Constellation Schema (Galaxy)

Star Schema

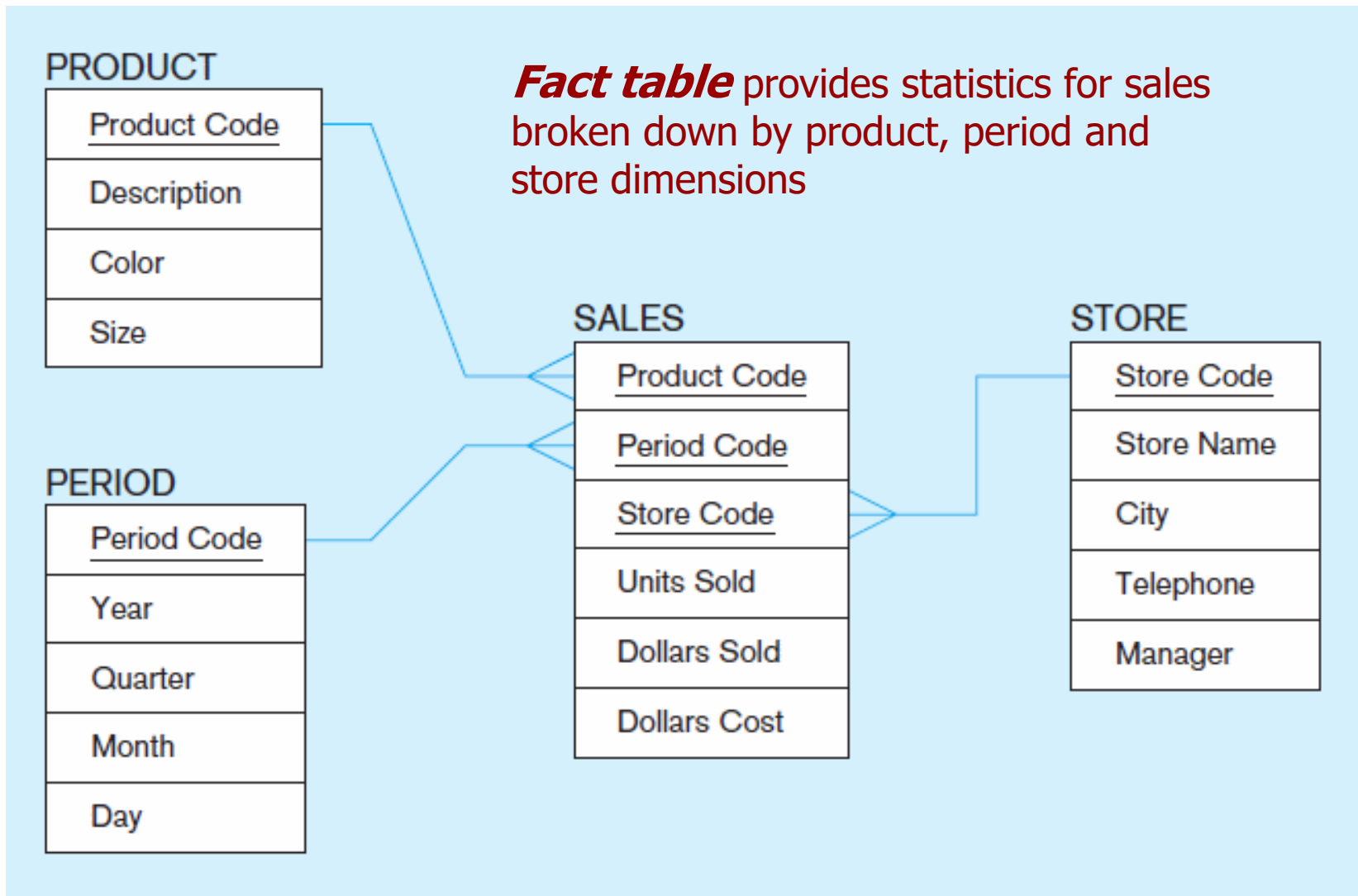
- A single, large and central fact table and one table for each dimension.
- Every fact points to one tuple in each of the dimensions and has additional attributes.
- Does not capture hierarchies directly.
- **Benefits:** Easy to understand, easy to define hierarchies, reduces no. of physical joins.

Components of a **star schema**



Excellent for ad-hoc queries, but bad for online transaction processing

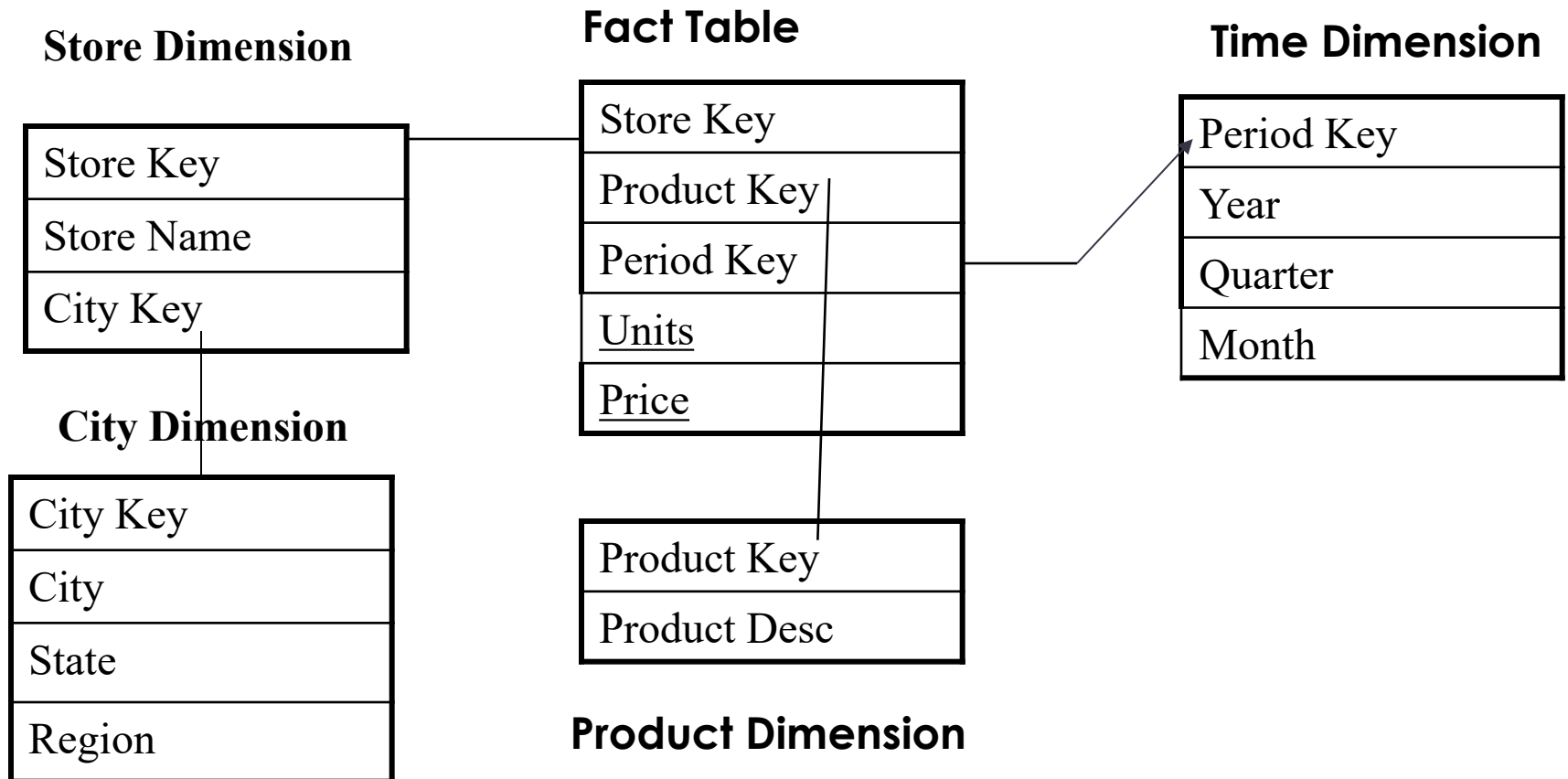
Star schema example



Snowflake Schema

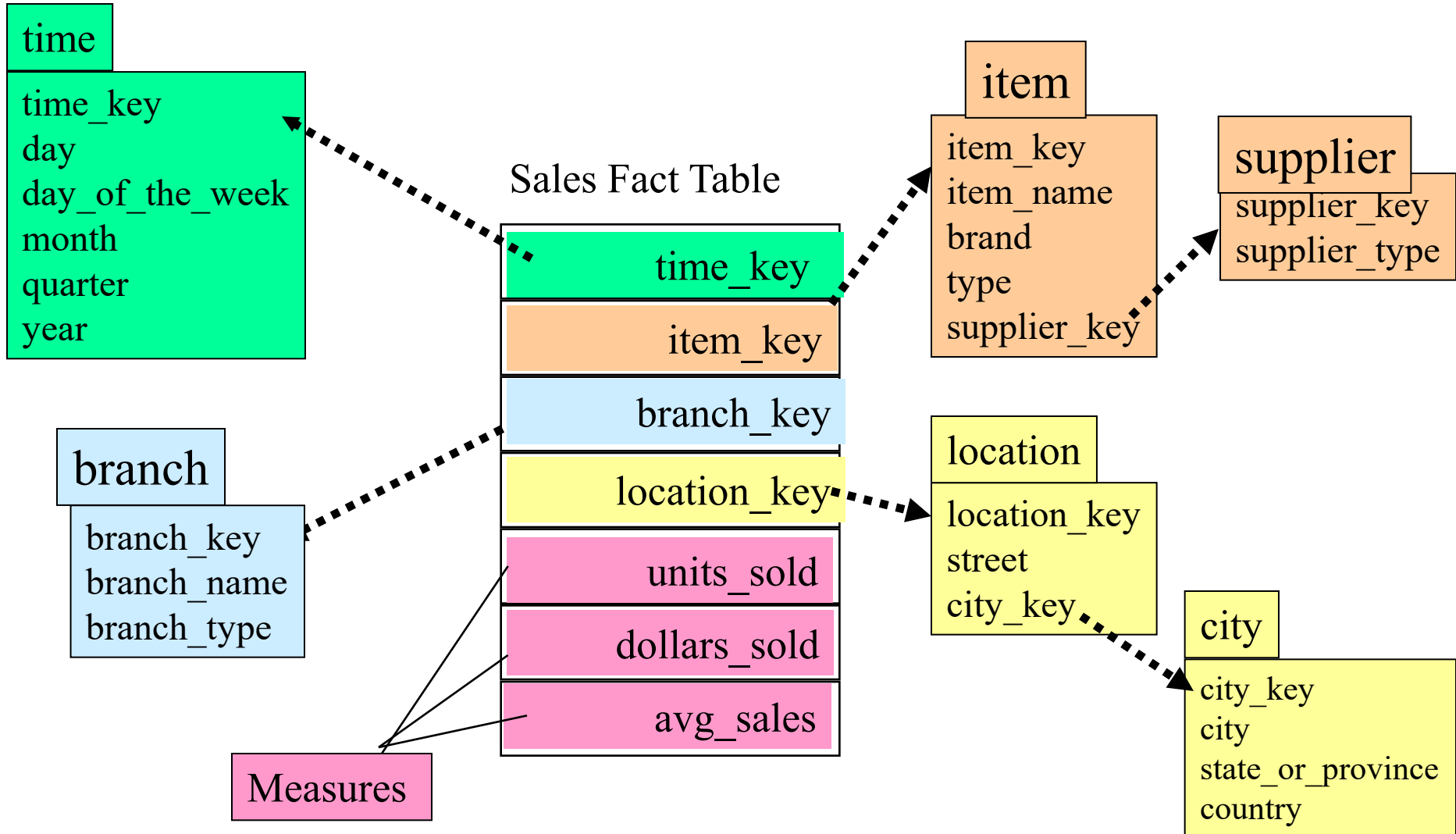
- Variant of star schema model.
- A single, large and central fact table and one or more tables for each dimension.
- Dimension tables are normalized i.e. split dimension table data into additional tables

Snowflake Schema (contd..)



Drawbacks: Time consuming joins, report generation slow

Example of Snowflake Schema



Fact Constellation(Galaxy Schema)

- Multiple fact tables share dimension tables.
- This schema is viewed as collection of stars hence called galaxy schema or fact constellation.
- Sophisticated application requires such schema.

Example of Fact Constellation

